

GOVERNMENT OF INDIA // SOVEREIGN THERMAL SURVEILLANCE

SOVEREIGN AGRICULTURAL BIOMASS ACTIVITY REPORT

SEVERITY: ABNORMAL

EVENT: EVT-IN-HAR-0022

National Technical Research Organisation (NTRO) • MoEFCC • CPCB Oversight

2026-09-02 13:44 IST (2026-09-02 08:14 UTC)

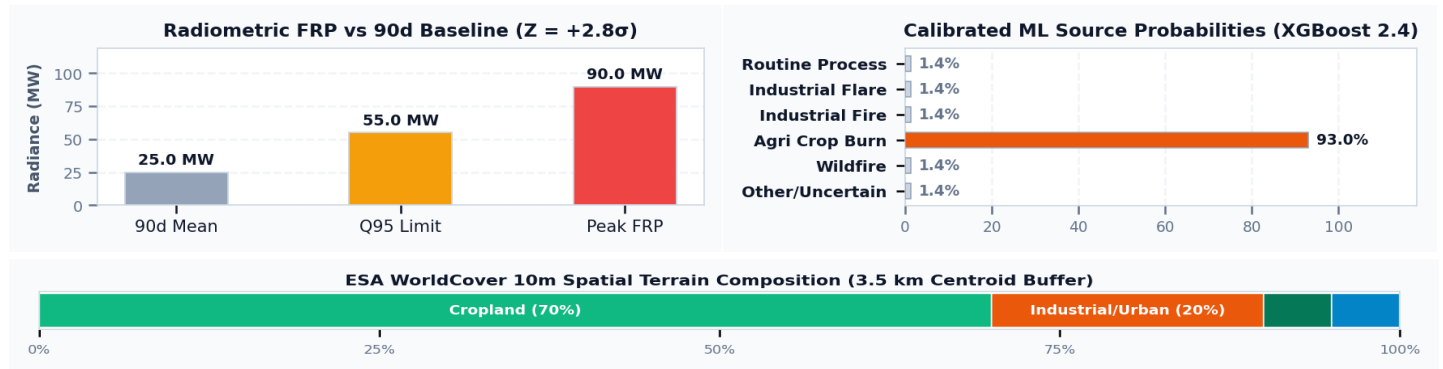
Thermal anomaly EVT-IN-HAR-0022 is classified as AGRI_BURN with 100.0% calibrated model confidence and severity tier ABNORMAL ($Z = +2.8\sigma$). Peak radiative output reached 90.00 MW across 1 verified multi-sensor satellite passes. Centroid mapped over Unknown without direct plant boundary overlap.

90.0 MW PEAK RADIANCE	1 Passes SATELLITE DETECTIONS	100.0% CALIBRATED SOFTMAX	+2.8σ ANOMALY DEVIATION
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1. Verified Multi-Sensor Radiometric Metrics

Metric Name	Observed Telemetry	Historical 90d Baseline	Radiometric Assessment
Peak Radiative Power (FRP)	90.00 MW	25.00 MW (Facility Mean)	3.6x Baseline Mean
Mean Cluster Radiance	65.00 MW	Persistence: TRANSIENT	Z-Score: +2.8σ Statistical Departure
Max Brightness Temp (BT)	320.0 K	Sensor Bands: VIIRS I4 (375m) / M13	High-Intensity Combustion Confirmed
Temporal Detection Window	2026-09-02T02:46 to 04:46 UTC	1 Sensor Passes Ingested	Active Multi-Sensor Track

2. Visual Radiometric Analytics & ML Attribution (High-DPI)



3. Centroid Location & Associated Industrial Plant Audit

Attribute	Forensic Detail	Attribute	Forensic Detail
Centroid Coordinates	29.50000°N, 76.30000°E	Primary Land Use	Unknown
State / Territory	Odisha	District / Jurisdiction	Angul
Associated Facility	No Direct Single-Facility Overlap	Spatial Buffer	Open Industrial / Rural Corridor

4. Nearby Sovereign Industrial Infrastructure (50 km Buffer)

Facility Name	Sector Category	State / District	Radial Distance	Baseline FRP
IOCL Panipat Naphtha Cracker Captive cogeneration power plant	Oil & Gas	Haryana	55.8 km	0.0 MW
IOCL Panipat cogeneration power plant	Oil & Gas	Haryana	55.9 km	0.0 MW
Industrial Facility	Nuclear	Haryana	65.1 km	0.0 MW
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5. Regional Active Anomaly Cluster & Concurrent Events (75 km Buffer)

Event Ref	Classification	Severity Tier	Peak Radiance	Distance	Latest Detection (UTC)
EVT-2026-7F253B	IND_ROUTINE	NORMAL	7.3 MW	54.1 km	2026-08-31 20:38
EVT-2026-61F3CC	IND_ROUTINE	NORMAL	2.5 MW	54.8 km	2026-08-27 21:55
EVT-2026-90AE4C	IND_ROUTINE	NORMAL	7.7 MW	55.1 km	2026-08-27 21:18
EVT-2026-56DFC7	IND_ROUTINE	NORMAL	5.2 MW	57.4 km	2026-08-31 20:57

6. Multi-Sensor Satellite Telemetry & Radiometric Pass Register

#	Detection Time (UTC)	Sensor Platform	Radiant Output	Brightness (K)	Day/Night
1	2026-09-02T02:46	VIIRS SNPP NRT (375m)	90.00 MW	320.0 K	Direct Telemetry Ingest

7. Statutory SOP Compliance Directives & Containment Protocol

ACTION 01: Immediate On-Site Physical Inspection: Dispatch SPCB / Regional Disaster Response team to verify combustion source and evaluate containment perimeter.
ACTION 02: Continuous Emission Telemetry (CEMS) Audit: Cross-verify Continuous Emission Monitoring Systems data and industrial flaring logs against satellite radiance timestamps.
ACTION 03: Safety Buffer & Containment Directive: Enforce active 2.5 km industrial safety perimeter and initiate cooling operations if thermal anomaly persistence exceeds 4.0σ.